Taiwan Semiconductor Manufacturing Co (TSM)

AI-Driven N2 Ramp Fuels Near-Term Surge, Valuation Still Stretched

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> TSMC's record-breaking quarters, led by a rapid 2-nanometer adoption and unprecedented high-performance-computing demand, demonstrate operating leverage that outpaces consensus expectations. Yet the market has priced the stock at a historic multiple that leaves little margin for error, especially given the capital intensity of the advanced-node transition. We therefore maintain a Hold rating with valuation concerns and a blended target of $242.72, implying a -43.2% upside from the current $427.50 price.

HOLD	$242.72	-43.2%	$427.50
Hold, Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market is fixated on short-term earnings beats, while the true story is a multi-year compounding of AI-driven demand, node transitions, and pricing power that will shape TSMC's trajectory through 2030.

EXECUTIVE SUMMARY

TSMC remains the world's preeminent pure-play semiconductor foundry, supplying leading-edge logic to the most powerful fabless designers. Advanced nodes of 7 nm and smaller account for 79 % of wafer revenue, while high-performance-computing (HPC) workloads represent 68 % of the mix. The company's monopoly position in sub-5 nm logic and its CoWoS advanced packaging give it a structural advantage that is difficult for competitors to replicate. The latest quarterly record underscores the strength of that position. Q3 FY2026 delivered $46.72 billion of revenue, a 41.2 % year-over-year increase, and a 68 % share of wafer revenue from the 2-nanometer node after a single-quarter jump from 3 % to 8 %. Gross margin softened to 66.4 % on early-stage N2 depreciation, but operating leverage pushed net profit to NT$763.06 billion, a 68.7 % rise YoY. The Q4 midpoint guide of $53.3 billion obliterates consensus and signals a continuation of the AI-driven surge. Capital allocation reflects a disciplined yet aggressive stance. FY2025 cash stood at $2,767.9 billion against total debt of $1,064.6 billion, leaving a net cash position of $1,703.3 billion. FY2026 capex guidance has risen to $60-64 billion, up from $40.9 billion in FY2025, to fund the 2-nanometer ramp and advanced packaging capacity. The balance sheet remains robust, supporting continued investment without jeopardizing liquidity.

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OPERATIONAL SNAPSHOT

$427.50	$3,809.1B	$992.4B	26.1%	1.37	$427.50
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-20.

THE MOAT IS DEEP AND DURABLE

TSMC’s moat rests on switching costs that are entrenched in the design-to-fab ecosystem. Fabless customers such as NVIDIA, AMD, Apple and Broadcom have co-engineered their most advanced products around TSMC’s process nodes, creating a lock-in that cannot be replicated without massive redesign. The company’s exclusive access to sub-5 nm logic and its CoWoS advanced packaging further raise the barrier to entry, as competitors lack both the lithography expertise and the integrated packaging know-how. Regulatory barriers reinforce the moat. The Section 232 semiconductor tariffs imposed in January 2026 and the evolving export-control posture on advanced computing ICs to China have heightened the strategic importance of TSMC’s Taiwan-based fabs. While these measures introduce geopolitical risk, they also limit the ability of rivals to source comparable capacity outside the region, preserving TSMC’s pricing power. Customer retention is evident in the accelerating adoption of the 2-nanometer node. The wafer mix leapt from 3 % to 8 % in a single quarter, a velocity unmatched in prior node transitions. This rapid migration signals confidence from hyperscalers and AI accelerator designers, who are willing to allocate substantial design wins to TSMC despite the early-stage margin drag. Pricing power is reflected in the consistent gross-margin expansion in the quarters preceding the N2 ramp. Q2 FY2026 posted a 67.7 % gross margin, well above guidance, while Q1 FY2026 achieved 66.2 % gross margin. Even as N2 depreciation pulled the Q3 margin to 66.4 %, management’s guidance of 65-67 % for Q4 suggests a temporary dip rather than a structural erosion.

VALUATION

Three scenarios make the valuation range explicit

The base case assumes a 15.0 % revenue CAGR from FY2026 to FY2030, with a terminal growth rate of 3.0 % and a target free-cash-flow (FCF) margin of 27.5 % in FY2030. The bear case trims revenue growth to 9.0 % and a target FCF margin of 27.2 %, while the bull case stretches revenue CAGR to 21.0 % and a target FCF margin of 27.8 %. All three scenarios are discounted at a weighted-average cost of capital of 9.89 %, reflecting an equity beta of 1.37 and a market risk premium embedded in the CAPM framework.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	9.0%	27.2%	$198.33	-53.6%

Base	50%	15.0%	27.5%	$241.02	-43.6%
Bull	25%	21.0%	27.8%	$290.52	-32.0%

Bear $198.33

The bear outcome is triggered by a prolonged margin drag from the 2-nanometer ramp, coupled with a slowdown in HPC demand that reduces the 68 % HPC share. If the 2-nanometer mix stalls below 8 % and capex intensity suppresses free-cash-flow conversion, the model projects a 9.0 % revenue CAGR and a 27.2 % target FCF margin, delivering a $198.33 price.

Base $241.02

The base path reflects the most likely trajectory, where the 2-nanometer node continues its rapid adoption, HPC demand remains robust, and margin compression is temporary. Under a 15.0 % revenue CAGR and a 27.5 % target FCF margin, the model arrives at a $241.02 price.

Bull $290.52

The bull case hinges on an accelerated AI supercycle that pushes HPC revenue share above 70 % and drives the 2-nanometer mix beyond 10 % of wafer revenue. Combined with sustained pricing power, a 21.0 % revenue CAGR and a 27.8 % target FCF margin lift the valuation to $290.52.

VALUATION FRAMEWORK

A live discount rate, not a static hurdle

Our discount rate of 9.89 % is derived from a CAPM construction that incorporates the model equity beta of 1.37, the risk-free rate, and the market risk premium. This rate reflects the market's pricing of TSMC's systematic risk, including the geopolitical exposure of its Taiwan-centric manufacturing base and the technology-cycle volatility inherent in advanced-node transitions. By using a live WACC rather than a static hurdle, we capture the dynamic cost of capital that evolves with macro-economic conditions and the firm's own risk profile. The 9.89 % WACC also embeds the premium investors demand for exposure to the AI-driven growth narrative. As the AI accelerator market expands, the implied equity risk may compress, but the current rate remains appropriate given the capital-intensive nature of the 2-nanometer ramp and the regulatory environment.

FCF margin normalization is the engine

Across the three scenarios, the target FCF margin moves modestly from 27.2 % in the bear case to 27.8 % in the bull case. The normalization process starts from the FY2025 free-cash-flow margin of 26.1 % and incorporates the expected margin uplift as the 2-nanometer node matures. Early-stage depreciation and capex intensity depress the Q3 FY2026 gross margin to 66.4 %, but historical

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2030**
Live WACC	**9.89% (2026-10-20)**
Terminal growth	**3.0%**
Equity beta	**1.37**
Diluted shares	**5.19 billion**

HOLD

over $242.72 blended target

patterns suggest that once the node reaches scale, gross margins will rebound toward the 67-68 % range observed in Q2 FY2026. Operating leverage, reflected in the FY2025 operating margin of 50.8 % and the FY2025 EBIT of $1,936.1 billion, provides a cushion that translates higher revenue into proportionally higher cash flow. The model therefore assumes a gradual improvement in FCF conversion as the advanced-node mix deepens, delivering the target margins used in the valuation.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

Methodology. Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

BUSINESS & SEGMENT DRIVERS

Revenue prospects anchor on AI-accelerator demand and 2-nanometer adoption HOLD

TSMC’s revenue engine is powered by two interlocking drivers. First, AI-accelerator demand from hyperscalers fuels the HPC segment, which already accounts for 68 % of platform revenue in Q3 FY2026 and grew 22 % sequentially. Second, the rapid adoption of the 2-nanometer node expands the advanced-node mix, raising the average selling price per wafer and deepening the technology moat. The company’s product mix is heavily weighted toward advanced logic. Wafer revenue from 7 nm and smaller nodes stands at 79 % across the latest quarters, while the 2-nanometer node alone contributed 8 % of wafer revenue in Q3 FY2026. This concentration on cutting-edge processes enables TSMC to command premium pricing and maintain high gross margins, even as early-stage depreciation temporarily compresses margins. The smartphone segment, while still representing roughly 21-22 % of revenue, has receded relative to HPC, reducing exposure to consumer-cycle volatility. This shift underscores the structural transition toward AI-centric workloads, a trend that management has highlighted through raised AI accelerator revenue CAGR guidance to 54-56 % for 2024-2029. What to monitor

- **2-nanometer ramp velocity** The leap from 3 % to 8 % wafer mix in a single quarter signals strong customer commitment; any slowdown could re-ignite margin concerns.
- **Capex execution** The $40.64 billion of capex through nine months of FY2026 tests TSMC’s ability to fund the node transition without eroding free-cash-flow conversion.
- **Geopolitical developments** Section 232 tariffs and export-control policies remain fluid; adverse changes could constrain demand from key AI customers.

- **HPC demand sustainability** The AI supercycle underpins the 68 % HPC share; a deceleration in hyperscaler spending would pressure revenue growth.

+31.6%	59.9%	50.8%	26.1%	$1,703.3B	31.5%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET CASH	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	$3,809.1B
Gross Profit	$2,281.3B
EBIT	$1,936.1B
Net Income	$1,697.6B
EBITDA	$2,742.1B
Free Cash Flow	$992.4B
Diluted EPS	$331.25
ROE	31.5%

RISK & MOAT MATRIX

The bear case

SHARE EROSION

Accelerated margin drag from the 2-nanometer ramp could depress earnings per share, prompting investors to rotate toward higher-margin peers. The Q3 gross margin of

M&A DIGESTION

While TSMC has not announced major acquisitions, any future M&A aimed at capacity expansion would need to be integrated efficiently. Missteps could strain

REGULATORY AND PRICING SCRUTINY

Section 232 tariffs and export-control uncertainties create a regulatory headwind that could limit sales to AI customers in China, reducing the effective addressable market for

TECHNOLOGY DISRUPTION

A breakthrough in alternative semiconductor technologies, such as silicon-photonic or quantum computing, could diminish the relevance of advanced silicon logic, threatening the long-term demand for

REINVESTMENT TREADMILL

The $60-64 billion capex range for FY2026 underscores a reinvestment treadmill that may compress free-cash-flow conversion for several years, limiting shareholder return potential.

WACC AND DURATION SENSITIVITY

Given the 9.89 % discount rate, any upward shift in market risk premia or beta would increase the cost of capital, reducing present value and widening the gap between

66.4 % already reflects early-stage depreciation, and any prolonged dip would erode confidence.

the balance sheet and distract from the node transition.

advanced nodes.

sub-5 nm wafers.

market price and intrinsic value.

The strengths

OLIGOPOLY POSITION

TSMC's dominance in sub-5 nm logic and its exclusive CoWoS packaging create an oligopolistic environment where few rivals can compete on technology or scale, reinforcing pricing power.

FCF COMPOUNDING

Operating margins of 50.8 % and a free-cash-flow margin of 26.1 % in FY2025 demonstrate the ability to convert high-margin revenue into cash, a foundation for sustained shareholder value.

GEOGRAPHIC DURABILITY

The concentration of fabs in Taiwan, coupled with strategic government support, provides a stable manufacturing base that is insulated from many global supply-chain disruptions.

PIPELINE OPTIONALITY

The ongoing 2-nanometer ramp and advanced packaging initiatives give TSMC multiple growth levers, allowing it to capture incremental premium pricing across several technology frontiers.

The bottom line on risk and reward

Our analysis finds that TSMC's operational fundamentals are exceptionally strong, with record revenue growth, a rapidly expanding advanced-node mix, and a balance sheet that comfortably supports aggressive capex. However, the market has priced the stock at a P/E of 1.3 x and an EV/revenue multiple of 0.1 x, leaving limited upside even if the AI supercycle continues unabated. The blended target of $242.72 reflects a -43.2 % downside from the current $427.50 price, underscoring valuation concerns that outweigh the near-term bullish catalysts. Accordingly, we assign a Hold rating with valuation concerns. Investors should monitor the velocity of the 2-nanometer ramp, capex execution, and geopolitical developments, as these factors will determine whether the stock can close the valuation gap or remain pressured.

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FRANCHISE DEEP DIVE

Advanced Logic Nodes: Sub-5 nm leadership

TSMC's flagship franchise is its sub-5 nm logic capability, which underpins the majority of AI accelerator and high-performance-computing workloads. The 7 nm and smaller node mix accounts for 79 % of wafer revenue, while

the 2-nanometer node already represents 8 % after a single-quarter surge. This deep penetration enables TSMC to command premium pricing and maintain gross margins in the high-60 % range, even as early-stage depreciation temporarily depresses margins. The 2-nanometer node’s rapid adoption is a direct response to hyperscaler demand for AI training and inference chips. Management’s Q4 FY2026 guide of $53.3 billion, well above consensus, reflects a backlog that is being filled by customers migrating to the leading node faster than any prior transition. The node’s scaling benefits also feed into the CoWoS advanced packaging platform, delivering higher integration density and performance for AI workloads. What the recent record shows

- Q3 FY2026 revenue of $46.72 billion, a 41.2 % YoY increase, driven by a 22 % sequential rise in HPC revenue share to 68 %.
- 2-nanometer wafer mix leapt from 3 % to 8 % of wafer revenue in one quarter, indicating accelerated customer migration.
- Gross margin compressed to 66.4 % on early-stage N2 depreciation, with management guiding Q4 gross margin between 65 % and 67 %.
- Capex through nine months reached $40.64 billion, underscoring the capital intensity of the node transition.

Our read: The sub-5 nm franchise is delivering the revenue acceleration and pricing power that justify the premium valuation, but the temporary margin drag and massive capex outlays create a valuation gap that we view as material. Until the market re-prices this gap, we maintain a Hold stance.